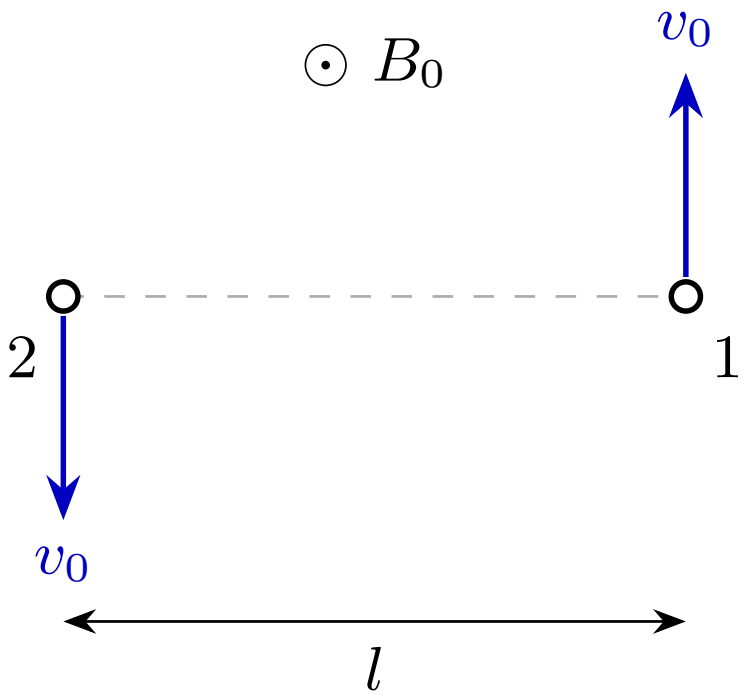
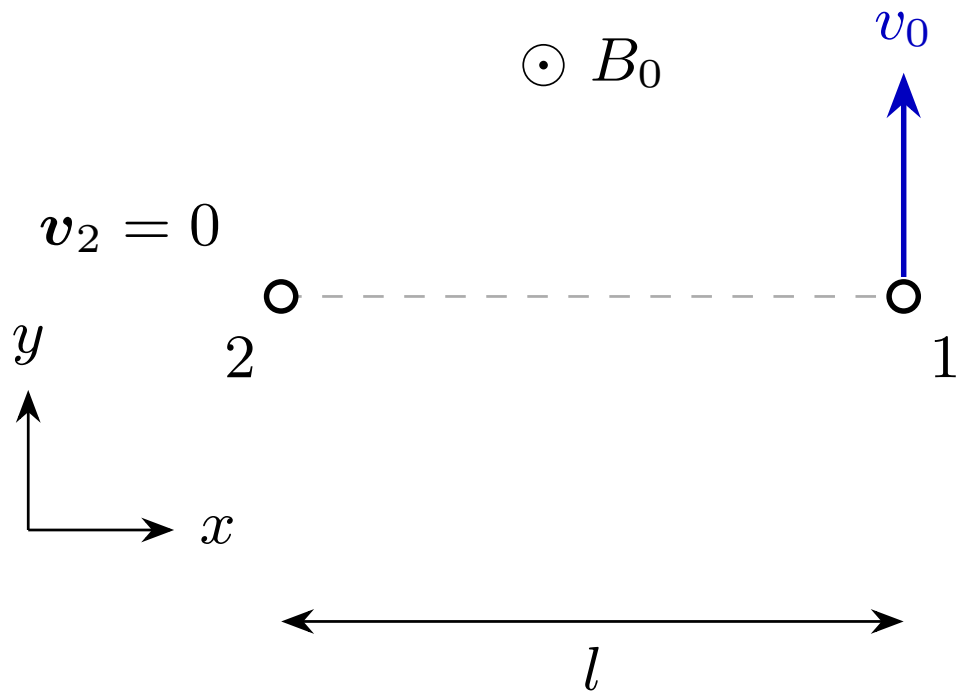
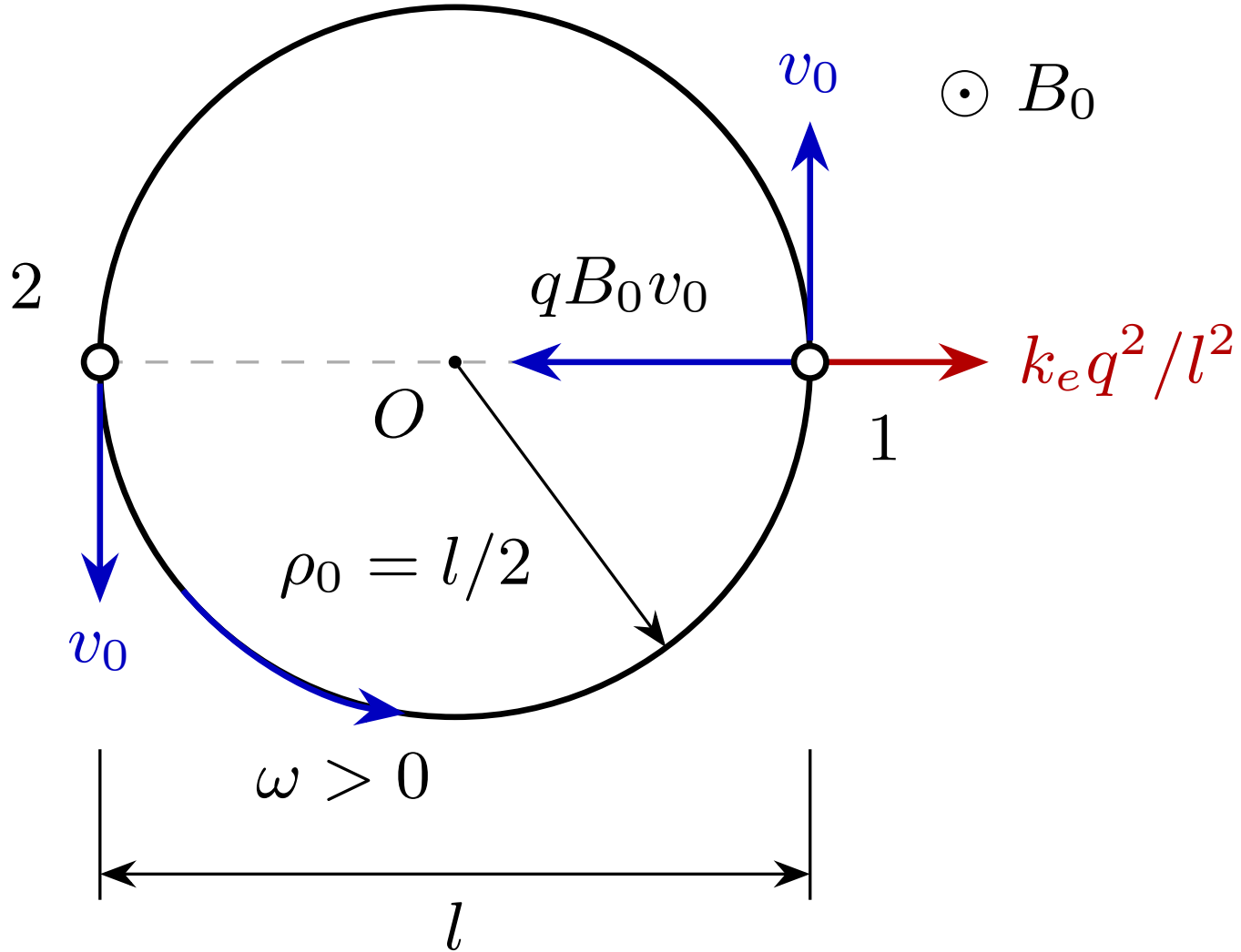
（1）等大反向

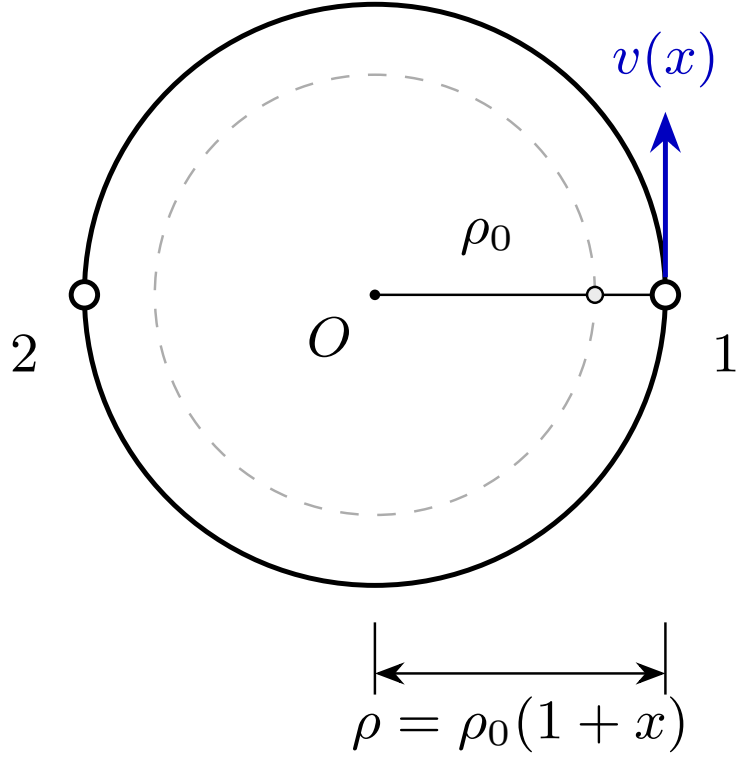


（2）仅电子 1 初始运动

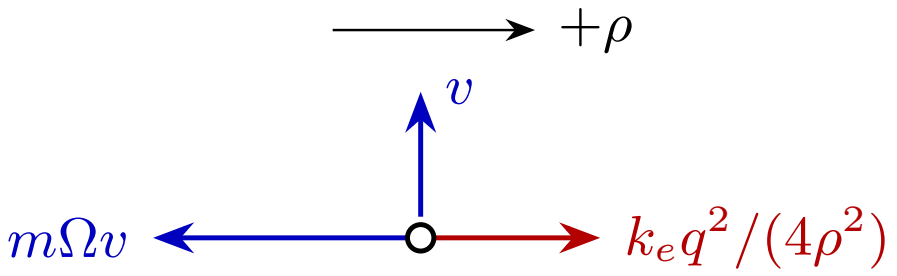



v_0
$\odot\ B_0$
2
qB_0v_0
k_eq^2/l^2
O
1
$\rho_0 = l/2$
v_0
$\omega > 0$
l

虚线：未扰动圆；实线：$\rho > \rho_0$



电子 1 的径向方程

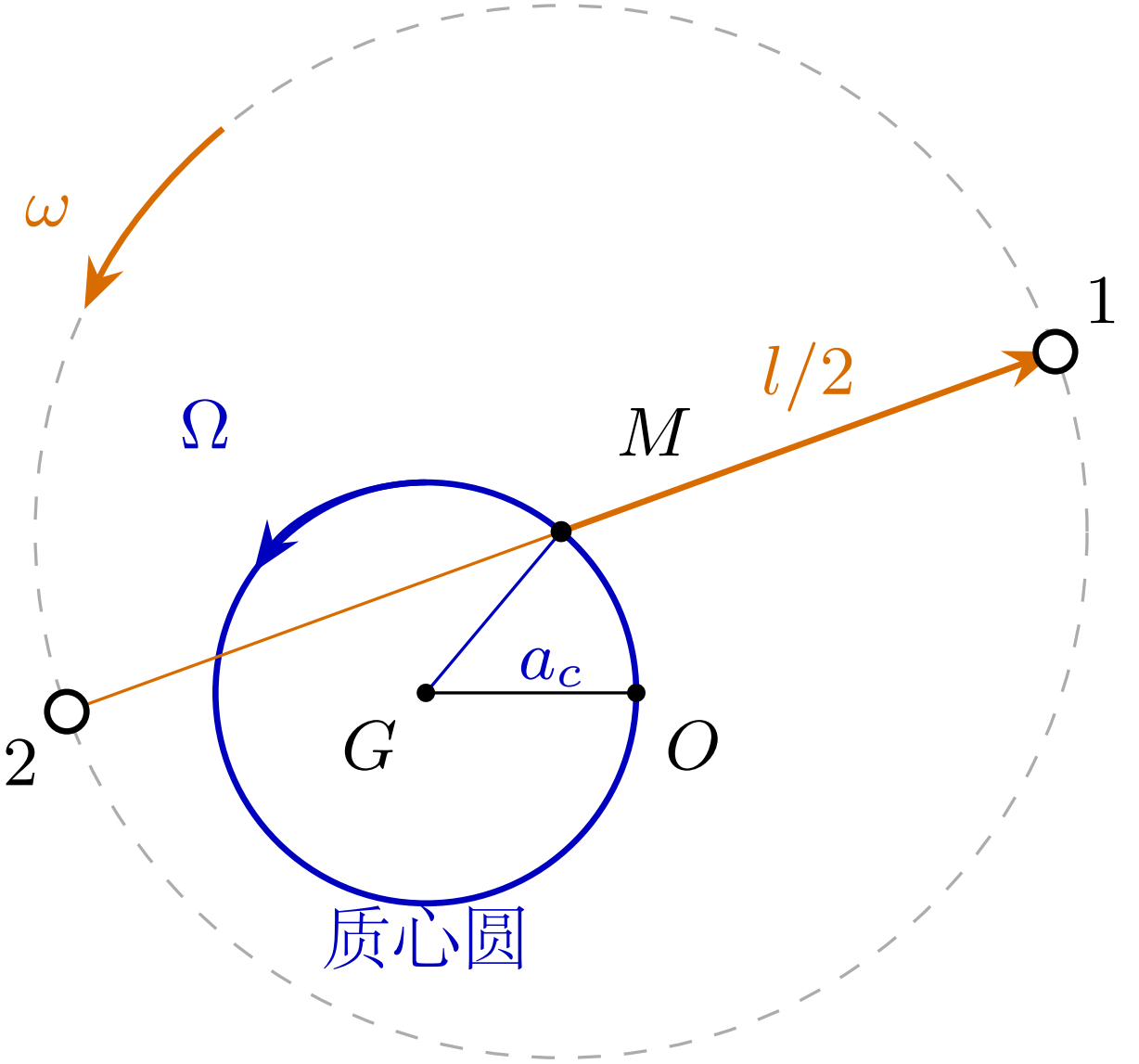


$$m\ddot{\rho} = mv^2/\rho + F_{\mathrm{e}} - F_{\mathrm{B}}$$

mv^2/ρ 来自加速度移项

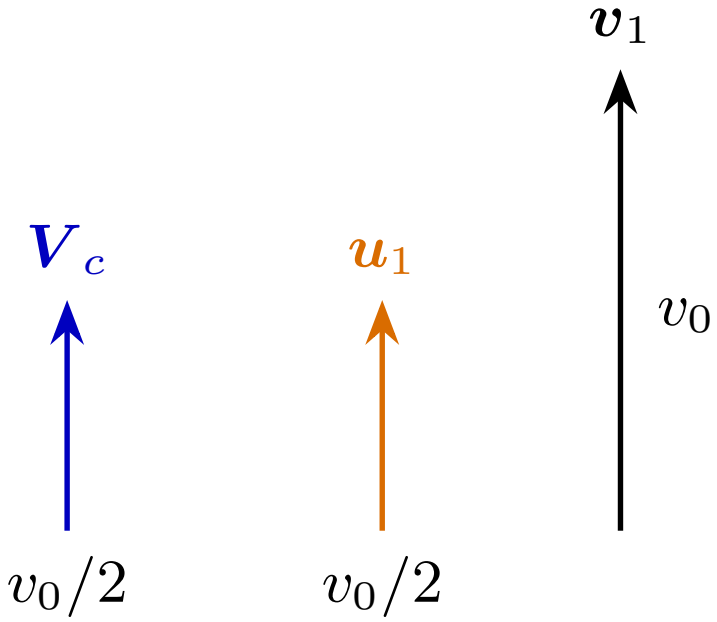
不是额外相互作用

质心绕 G 转，电子绕当前质心转



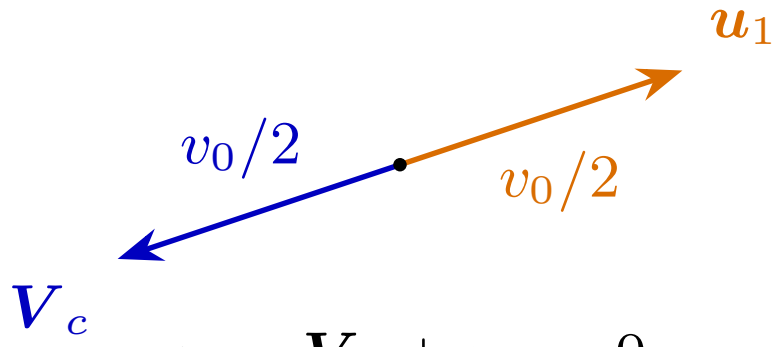
$a_c = v_0/(2\Omega)$；橙色连线长度为 l

两速度分量同向



$t=0$，相位差为 0

首次相消；两分量等模反向



$$\boldsymbol{v}_1 = \boldsymbol{V}_c + \boldsymbol{u}_1 = 0$$

$$(\Omega - \omega)t_* = \pi$$